The Paleomagnetism of a Triassic Diabase Dike in Nova Scotia

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The paleomagnetism of a 110-km-long diabase dike in the southern part of Nova Scotia has been studied. A potassium-argon whole-rock age determination has confirmed the Triassic age assignment to the dike rocks. The magnetic stability and other magnetic properties of the rocks were examined, and the presence of a stable component was established. An analysis of the paleomagnetic data establishes their statistical significance and indicates that at the time of injection of the dike the North Pole was situated in the area presently occupied by northern Siberia (98°E, 69°N). This pole position is in good agreement with available paleomagnetic poles obtained for other North American Triassic formations.

INTRODUCTION

The paleomagnetism of several North American Triassic formations has been studied during the past ten years. However, it is generally considered today that much of the available data are of questionable reliability, and a specific selection was proposed five years ago by *Opdyke* [1961] in a review of the subject. This paper provides new data obtained for a suite of oriented samples collected from two localities (Figure 1), 70 km apart, along a 110-km-long diabase dike that spans the southernmost part of Nova Scotia. In addition, a K-Ar whole-rock age was determined for material from one of the sampling localities.

GENERAL GEOLOGY AND AGE OF THE DIKE

Faribault [1912], who reported on the eastern part of the dike, described it as '. . . an important dyke of diabase, 200 to 500 feet wide, extending along the sea-shore in a westward direction for at least 25 miles from west Ironbound Island . . . to Liverpool.' *Taylor* [1960] stated that whereas the diabase crops out at only a few localities, the dike extends in a west-southwest direction for a distance of 110 km and that boulders of this rock type are frequently found along the strike of the dike. On aeromagnetic maps published by the Geological Survey of Canada (Maps G-608, -609, -612, -613) the magnetic expression of the dike is shown as a prominent feature throughout its total length.

The dike rocks cut a granite mass north of Shelburne [*Taylor*, 1960] that has been dated at 315 m.y. [*Fairbairn et al.*, 1960]. A K-Ar age was determined on a sample of whole rock collected at site B (Figure 1). The results of this analysis are

K	= 0.71%
Ar^{40}/K^{40}	= 0.01197
Radiogenic Ar^{40}	= 40%
Age[1]	= 197 ± 32 m.y.

(Error calculated at 95% confidence level.)

λ_e	= 0.585×10^{-10} yr^{-1}
λ_{total}	= 5.30×10^{-10} yr^{-1}

NATURAL REMANENT MAGNETIZATION

Eleven separately oriented cores were collected with a portable diamond drill. The cores, 2.54 cm in diameter, were oriented in situ with a solar compass and a sampling tube of the type described by *Gough and Opdyke* [1963]. Two cylinders 2.25 cm long were cut from each core, and the intensity and direction of magnetization of both were measured with a 3-magnet automatic astatic magnetometer in the laboratories of the Geological Survey of Canada. The vector sum of the two unit vectors representing the magnetization of each cylinder of a given core was calculated and taken as the mean direction of magnetization for that core. The maximum divergence within individual

[1] Error calculated at 95% confidence limit. See *Wanless et al.* [1965] for details on calculations and analytical procedures. $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}; $\lambda_{total} = 5.30 \times 10^{-10}$ yr^{-1}.

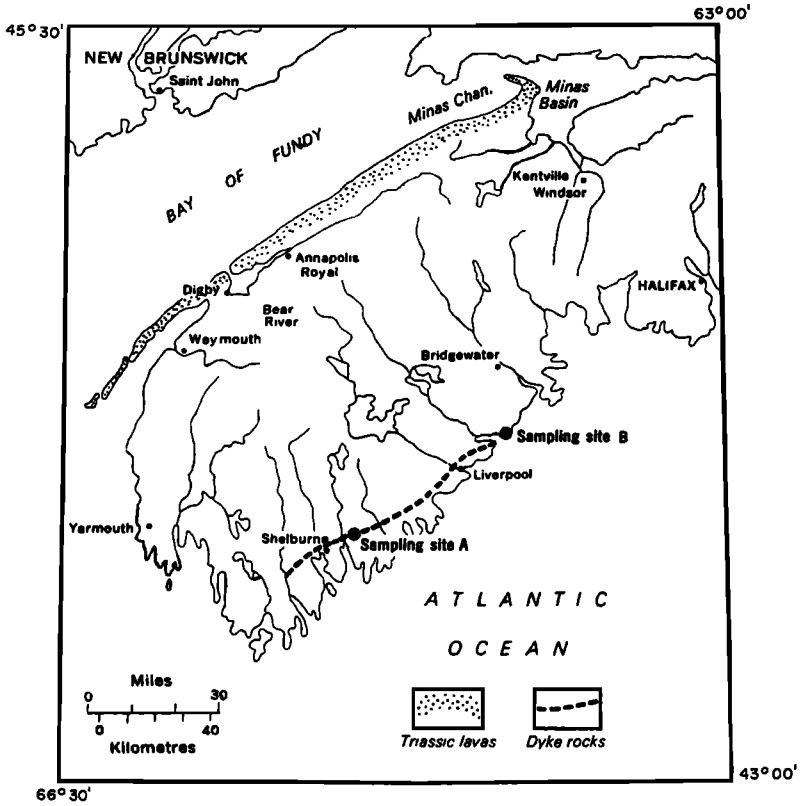


Fig. 1. Generalized map of the area.

pairs of vectors was 6°. The mean directions are plotted stereographically in Figure 2. Table 1 presents statistics computed from three *Fisher* [1953] analyses of the directions, based on the following populations: (*a*) the eleven individual samples; (*b*) the samples from each site considered as a group, and (*c*) the mean directions calculated for the samples at each site.

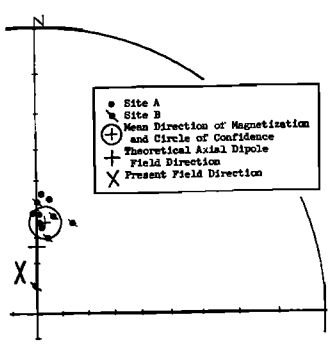


Fig. 2. Stereographic projection of natural remanent magnetization directions.

Magnetic Stability

Two cores, one from each site, were tested for magnetic stability. The two cylinders of each core were progressively demagnetized in a 60-cps alternating field of decreasing amplitude [*Larochelle and Black*, 1965]. For the

TABLE 1. Fisher Statistics for Directions of Natural Remanent Magnetization

Population	*D*	*I*	*N*	*R*	*k*	α_{95}
Eleven samples	4.9	54.0	11	10.843	63.8	5.7
Site A	2.4	50.6	6	5.981	265.2	4.1
Site B	8.5	58.1	5	4.890	36.4	12.8
A and B	5.2	54.4	2	1.995	193.4	18.1

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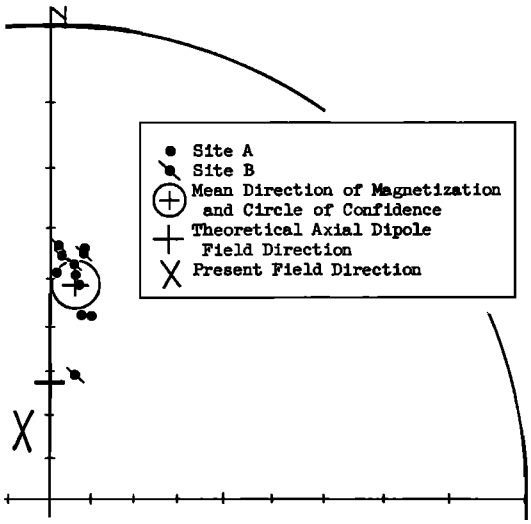


Fig. 3. Stereographic projection of directions of remanent magnetization after alternating field treatment (350-oe peak).

first treatment the peak amplitude of the field was first set at 100 oe, and for the successive treatments it was raised by steps of 50 oe to a maximum of 600 oe. Magnetization was measured after each treatment. For all treatments the axis of the first cylinder was initially oriented along the positive axis of the demagnetizing coil and that of the second specimen was oriented in the opposite direction. The elimination of a soft viscous magnetization component was noted after the first treatments at low peak amplitudes. This component was practically eliminated after the treatment at 300 oe, although the thermoremanent and more stable component was then also reduced appreciably. After the treatment at peak field (600 oe) the magnetization of two of the specimens was reduced to below 10^{-7} cgs units/cm^3 and could not be measured accurately.

One cylinder of each of these two cores was used to determine the saturation field, H_{sat}, the saturation magnetization, J_{max}, and the coercivity of maximum isothermal remanence, H_{cr}. The values obtained for these parameters were identical for both specimens and were 1500 oe, 0.037 cgs units/cm^3, and 375 oe, respectively. They are diagnostic of a relatively high magnetic stability for rocks of this type.

Remanent Magnetization After Magnetic 'Cleaning'

In the light of the above information, the remaining nine samples were 'cleaned' in an alternating field of 350 oe peak amplitude. This treatment completely eliminated the viscous isothermal component, while leaving enough of the thermoremanent component to permit accurate measurement. The mean directions are plotted in Figure 3. Three Fisher analyses of these directions were made at the population levels mentioned earlier. Values of 978 and 86 were calculated for estimates of the between- and within-site precision indices b and w, respectively [*Watson*, 1956]. Since the ratio w/b is less than the statistic $F_{2,18}$ at the 95% probability level, the individual site mean directions are considered to be identical and the eleven samples are considered to have a common mean. Accordingly, the mean direction of the eleven samples was used in computing the virtual pole and the axes of the corresponding ellipse of confidence. The results are listed in Table 2.

The rocks dealt with have a relatively high magnetic stability; the mean directions obtained for the two widely spaced sampling localities are in excellent agreement, and they are significantly distinct from that of the present-day field in the general area. The values obtained for precision indices k, w, and b would attest to the reliability of the data and indicate that more extensive sampling would not significantly affect the result.

Comparison with Data on Other Formations and Conclusions

Comparing the new pole position with those obtained by other workers for various Triassic

TABLE 2. Fisher Statistics for Directions of Remanent Magnetization after Treatment in Alternating Field of Peak Amplitude 350 Oersteds

Population	D	I	N	R	k	α_{95}
Eleven samples	6.6	40.8	11	10.894	94.4	4.7
Site A	7.5	41.1	6	5.975	195.8	4.8
Site B	5.5	40.5	5	4.921	50.4	10.9
A and B	6.5	40.8	2	1.999	5276.4	3.4

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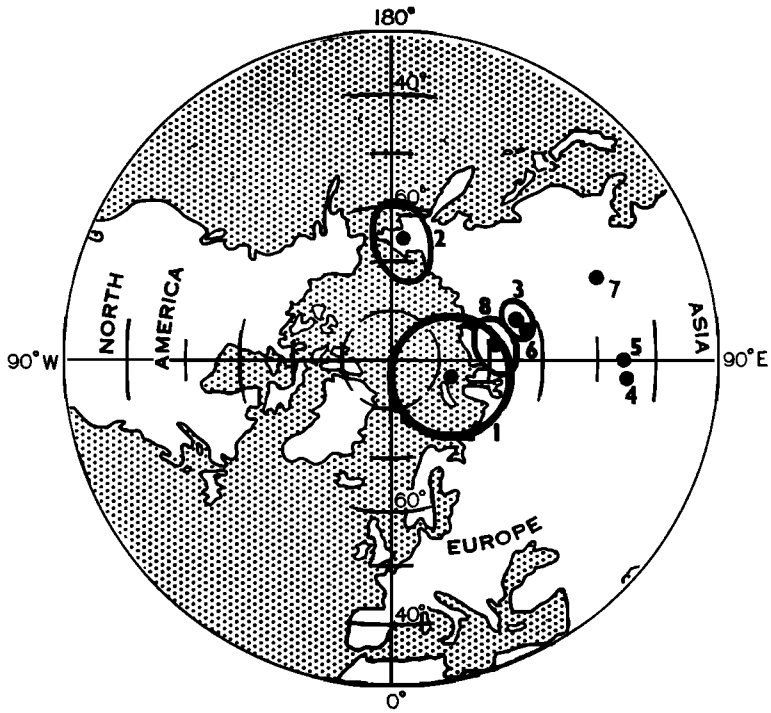


Fig. 4. Pole positions derived from various North American Triassic formations.

North American formations is of interest. *Opdyke* [1961] regarded as acceptable the results of *Collinson and Runcorn* [1960] on the Moenkopi and the Chugwater formations, those of *DuBois et al.* [1957] on the Connecticut Valley rocks, those of *Irving and Banks* [1961] on the Massachusetts lavas, and his own on the Newark series. On the other hand, he did not consider Bowker's data [*Cox and Doell,* 1960] on the Nova Scotia Triassic lavas—not to be confused with the dike discussed in this paper—and *Graham's* [1955] data on the New Oxford sediments as useful for comparison. We consider that Opdyke's results on the Newark series are reliable. On the other hand, the accuracy of the poles derived from the results of DuBois et al., Irving and Banks, and Collinson and Runcorn is open to question because the mean directions used in the derivation of these pole positions and the associated statistics were computed by

TABLE 3. Pole Positions Derived for Various North American Triassic Formations

Formation	Pole Position		$\delta\chi$, deg	$\delta\psi$, deg	Investigators
	E Longitude, deg	N Latitude, deg			
1 Nova Scotia lavas	75½	77	11½	11½	Bowker
2 New Oxford sed.	174	66	8½	5½	Graham
3 Newark series	108	63	4	3	Opdyke
4 Connecticut rocks	86	54			DuBois et al.
5 Massachusetts lavas	90	55			Irving and Banks
6 Moenkopi	103	62			Collinson and Runcorn
7 Chugwater	112½	48			Collinson and Runcorn
8 Nova Scotia dike	98	69	5½	3½	Larochelle and Wanless

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taking the result for the specimen, rather than for the sample, as the basic independent observation. This practice may have the net result of giving too much weight to those samples represented by a large number of specimens. It also gives the false impression of a much greater precision than actually exists for a particular population. We believe Bowker's results should be considered provisionally as valid because their computation was based on the site means and because the natural remanent magnetization was cleaned by alternating field treatment. Finally, there is no apparent reason to reject Graham's data on the New Oxford sediments.

Figure 4 and Table 3 give the pole positions derived from the seven sets of data referred to above and the one based on the new data. Entries 1, 2, 4, 6, and 7 were extracted from Cox and Doell's review. The statistics for entries 4, 5, 6, and 7 are omitted for the reasons given above.

In general, these data indicate that during Triassic time the North Pole was located in the region occupied today by northern Siberia. Particular attention is drawn, however, to the data on the New Oxford sediments, which suggest that these rocks were magnetized during a different geological epoch. Assuming that it is established beyond doubt that the New Oxford beds were deposited in Triassic time, there remains the possibility that they acquired their present magnetization at a much later date. The pole position derived from these rocks is in fact in excellent agreement with those derived for Cretaceous rocks by *Larochelle* [1961], *Larochelle et al.* [1965], and *Currie et al.* [1963]. It would be of interest to study this possibility in greater detail.

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